WasteWise: An Explainable AI Decision Support Framework for Urban Waste Governance

Research Foundation (Independent Concept Document)

1. Problem Statement (Detailed)

Urban Local Bodies (ULBs) are responsible for collecting, transporting, processing and disposing of municipal waste. Although most cities have waste management rules, many municipalities still struggle because decisions are based on manual reports, fragmented information and delayed analysis rather than real-time intelligence.

- Waste generation changes daily due to population growth, festivals, tourism, weather and commercial activity, but planning often relies on old estimates.
- Waste segregation at source is inconsistent, making recycling and treatment less effective.
- Different departments maintain separate data, making it difficult to obtain a complete picture.
- Landfill capacity is monitored manually, so overflow risks are identified late.
- Officers receive reports but not clear recommendations on what action should be taken first.
- Most systems react after problems occur instead of predicting them in advance.
- The overall result is higher operating cost, inefficient resource allocation, environmental pollution and slower public service.

2. Current Existing Solutions (Detailed)

Many municipalities already use digital tools. These solutions improve individual activities but do not provide complete decision intelligence.

- **Swachh Bharat Mission** – Provides national guidelines and encourages scientific waste management.
- **GPS Vehicle Tracking** – Tracks garbage trucks but does not recommend better deployment.
- **Smart Bins and IoT Sensors** – Measure bin fill level but usually work independently.
- **Citizen Complaint Applications** – Allow residents to report issues but do not analyse city-wide patterns.
- **Municipal ERP Systems** – Store administrative records but offer limited prediction and analytics.
- **Material Recovery Facilities (MRFs)** – Separate recyclable materials but depend on proper segregation.
- **Composting and Bio-methanation** – Treat organic waste effectively but require proper planning and monitoring.
- **Landfills** – Safely dispose residual waste but require capacity planning and environmental monitoring.

3. Gaps That Still Exist

- **Static Planning** – Planning uses historical data instead of continuously changing real-world conditions.
- **No Integrated View** – GPS, complaints, landfill, weather and processing data remain disconnected.
- **Reactive Management** – Problems are handled only after overflow, complaints or pollution occur.
- **Limited Prediction** – Existing systems rarely forecast waste generation or landfill utilisation.

- **No Explainability** – Dashboards show numbers but not the reason behind recommendations.
- **Manual Decision Making** – Officials still depend on experience instead of AI-supported evidence.
- **Resource Inefficiency** – Vehicles, manpower and processing plants are not optimally allocated.

4. Proposed Solution – WasteWise

WasteWise is proposed as an Explainable AI Decision Support Framework that continuously assists municipal authorities in planning, monitoring and improving waste management operations.

- Collect data from GPS-enabled vehicles, IoT sensors, landfill records, citizen complaints, weather information and historical databases.
- Integrate all information into one centralized municipal knowledge platform.
- Apply Artificial Intelligence models to predict future waste generation, identify high-risk wards and estimate landfill capacity.
- Use Explainable AI (XAI) to clearly explain why every prediction or recommendation is generated.
- Prioritize actions based on urgency, available resources and expected impact.
- Display recommendations on an interactive dashboard for municipal officers.
- Continuously update predictions as new data becomes available.

Expected Benefits

- For Municipal Officers: Faster, evidence-based decisions and better planning.
- For Citizens: Cleaner neighbourhoods and quicker response to complaints.
- For Government: Better resource utilisation and improved transparency.

- For Environment: Reduced landfill overflow, improved segregation and lower pollution.

Glossary (Full Forms and One-Line Meanings)

Term	Full Form	Meaning
ULB	Urban Local Body	Municipal authority responsible for city administration.
MSW	Municipal Solid Waste	Daily waste generated from homes, markets and offices.
AI	Artificial Intelligence	Technology that learns from data to support decisions.
XAI	Explainable Artificial Intelligence	AI that explains why it produced a recommendation.
IoT	Internet of Things	Connected devices that collect and share real-time data.
GPS	Global Positioning System	Technology used to track vehicle locations.
MRF	Material Recovery Facility	Facility where recyclable waste is sorted and recovered.
ERP	Enterprise Resource Planning	Software used to manage organisational records.
DPR	Detailed Project Report	Planning document prepared before executing a project.
Dashboard	-	A visual interface showing important information and recommendations.